Do Better Captions Hallucinate Less?

Measuring Object Hallucination in CNN–LSTM Image Captioning with and without Attention

Sachin

October 2026

Abstract

Image-captioning systems are usually ranked by n-gram overlap, yet a caption can score well while naming objects that are not in the image. This project builds a controlled COCO benchmark with three captioners that share the same training data, features and decoder: a global-feature CNN–LSTM, the same model with Bahdanau attention, and an attention model that additionally receives detected object labels. All three are evaluated on 500 held-out COCO images with BLEU-1–4, CIDEr-D and CHAIR-style object-hallucination metrics. The BLEU leader (Object-aware) was also the model with the lowest object-hallucination rate (23.64%). The pooled caption-level association is a moderate **negative** correlation (Spearman ρ = -0.327; image-cluster $p < 0.001$): captions with higher sentence BLEU-4 tend to contain slightly fewer hallucinated objects. Across the three model-level operating points, the rank correlation between BLEU-4 and CHAIRs is -1.00. The results support a simple conclusion: BLEU and hallucination are related but not interchangeable. At matched BLEU levels the models still differ substantially in hallucination, so captioning evaluations should report both.

1. Introduction

Automatic image captioning has improved dramatically, but the dominant evaluation metrics — BLEU, METEOR, ROUGE and CIDEr — reward overlap with reference sentences. That objective is only loosely connected to visual grounding. A model can learn that "a man is standing on a street" is a safe, frequent caption and emit it for many images; such a caption may contain plausible objects that are absent from the specific image. This failure mode is called *object hallucination* and is invisible to a corpus-level BLEU score.

The assignment asks four questions:

1. Can a small CNN–LSTM captioner be trained on a dataset with captions and object annotations?
2. Does adding visual attention change caption quality and hallucination?
3. Does supplying detected object labels reduce hallucination?
4. Does better BLEU actually mean fewer hallucinations?

We answer them with a reproducible, laptop-scale experiment. The code, cached features, detector outputs, metrics and figures are included in the repository.

2. Related work

CNN–LSTM captioning. Show and Tell encodes an image with a CNN and decodes a sentence with an LSTM initialised from the image feature (Vinyals et al., 2015). Show, Attend and Tell adds a spatial attention mechanism that lets each generated word focus on different image regions (Xu et al., 2015). Bottom-up and top-down attention later replaced grid features with region proposals (Anderson et al., 2018).

Object hallucination. Rohrbach et al. (2018) showed that captioning models frequently mention objects that are not present, and introduced the CHAIR metrics used here: CHAIRs measures the fraction of mentioned objects that are hallucinated, while CHAIRi measures the fraction of captions with at least one hallucinated object. Their analysis also showed that standard caption metrics do not reliably detect this failure.

Metrics. BLEU is a clipped n-gram precision with a brevity penalty (Papineni et al., 2002). CIDEr-D scores consensus using TF-IDF weighted n-grams and a length penalty (Vedantam et al., 2015). COCO provides five reference

captions per image and 80 object categories (Lin et al., 2014).

3. Data

We use the COCO 2017 validation images because they contain both captions and instance annotations, and because the full 18 GB train2017 archive is a poor fit for a 16 GB laptop. The 5,000 images are split deterministically (seed 42) into **4,000 train**, **500 validation** and **500 test** images. The caption vocabulary is built only from the training split.

Split	Images	Captions
Train	4,000	20,010
Validation	500	2,503
Test	500	2,501

The vocabulary contains 2,916 entries (2,912 lexical word types plus four special tokens; minimum frequency 3) and covers 97.81% of training tokens. Training captions have a mean length of 10.4 tokens (maximum 39; the full corpus maximum is 49). Each image has 2.93 annotated COCO objects on average.

Ground-truth objects for CHAIR are the unique COCO category names in `instances_val2017.json`; 48 images have no annotations and therefore an empty ground-truth object set. Detected labels come from a COCO-pretrained Faster R-CNN with a MobileNetV3-320 backbone, run on the same images at a score threshold of 0.30 and at most 12 detections per image. The detector is independent of the captioners and never sees test labels: its output is the only object information available to the object-aware model.

4. Models

All models use a frozen ImageNet-pretrained ResNet-18. The `layer4` feature map is a 7×7 grid of 512-dimensional vectors; its global average is a 512-dimensional image vector. Features are extracted once and memory-mapped, so the LSTM never runs a CNN and the whole experiment fits in a few hundred MB.

Global (no attention). A two-layer projection initialises the LSTM hidden and cell states from the global feature. At each step the LSTM receives only the previous word embedding. This is the Show-and-Tell baseline.

Attention. A Bahdanau-style additive attention module scores the 49 spatial locations against the previous hidden state, producing a softmax map and a 512-dimensional context vector. The LSTM input is the concatenation of the word embedding and the context vector. Attention maps are stored during decoding for visualisation.

Object-aware. The attention model additionally receives the detected object labels. Each label is embedded (128 dimensions), weighted by its detector confidence, and averaged into one object vector. That vector is concatenated to the LSTM input at every time step. An image with no detections receives a zero vector. This design tests whether explicit object evidence reduces mentions of absent objects.

Training details. AdamW with a OneCycle schedule, learning rate 0.001, weight decay 0.0001, dropout 0.3, label smoothing 0.05, gradient clipping 5.0, batch size 96. Each epoch samples 8,000 caption–image pairs from the 20,010-pair training pool; models are trained for 12 epochs. Checkpoints are selected by validation loss, not by BLEU, to avoid selecting a degenerate high-BLEU model.

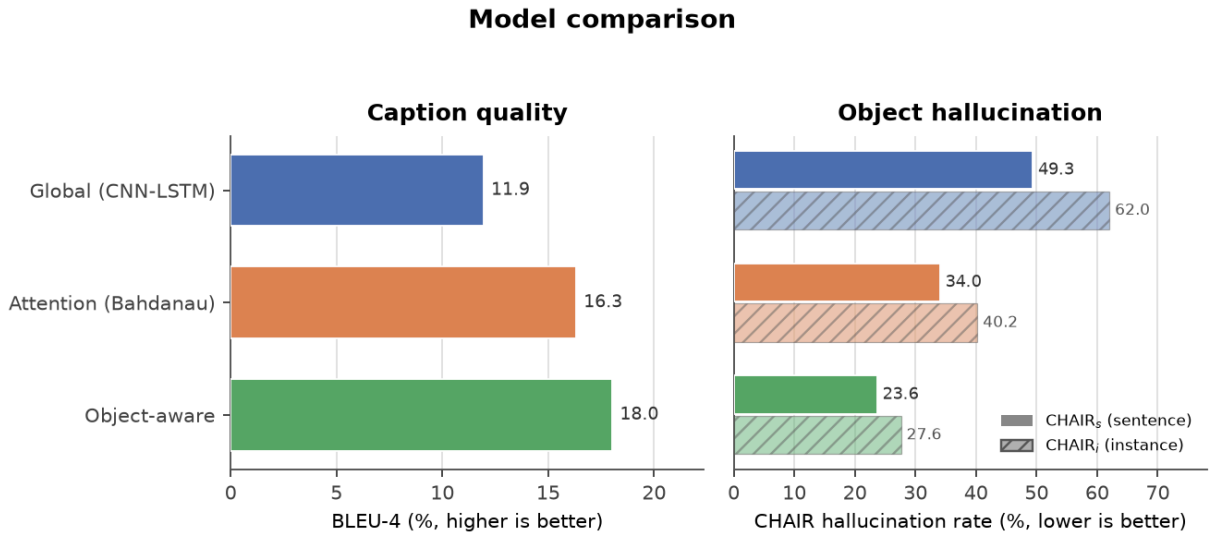
5. Evaluation

BLEU-1–4. SacreBLEU with the 13a tokenizer; BLEU-k is the cumulative score over 1…k grams. **CIDEr-D.** A from-scratch implementation with global document frequencies, TF-IDF cosine similarity and the CIDEr-D length penalty.

CHAIR. We map caption words to the 80 COCO classes using a curated synonym table (longest phrase first, so "hot dog" is not counted as "dog"). A mention is hallucinated if the class is absent from the image's instance annotations. CHAIRs is the number of hallucinated unique object mentions divided by all object mentions; CHAIRi is the fraction of images with at least one hallucinated mention. Non-COCO nouns such as sky, grass and water are ignored.

Statistics. Confidence intervals are percentile bootstraps over images (1,000 resamples). Paired comparisons resample the same image indices for both models. We also report a McNemar test for the paired binary "any hallucination" outcome and Spearman correlations between sentence BLEU and hallucination counts.

6. Results



Captioning quality and object hallucination for the three models. Lower CHAIR is better.

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr-D
Global	56.84	35.74	20.65	11.94	0.485
Attention	59.75	40.59	26.09	16.32	0.704
Object-aware	61.96	43.35	28.44	18.03	0.815

Table 1. Caption-quality metrics. BLEU-k is cumulative BLEU over 1…k grams; CIDEr-D is on a 0–10 scale.

Model	CHAIRs (%)	CHAIRi (%)	Object P (%)	Object R (%)	Object F1 (%)
Global	49.32	62.00	50.68	30.04	37.72
Attention	34.01	40.20	65.99	35.68	46.31
Object-aware	23.64	27.60	76.36	40.00	52.50

Table 2. Grounding metrics. Object precision/recall/F1 compare the COCO object mentions in the generated captions with the image's COCO annotations.

The best BLEU-4 is **18.03** (Object-aware), the best CIDEr-D is **0.815** (Object-aware), and the lowest CHAIRs is **23.64%** (Object-aware). The BLEU leader (Object-aware) was also the model with the lowest object-hallucination rate (23.64%).

6.1 Paired comparisons

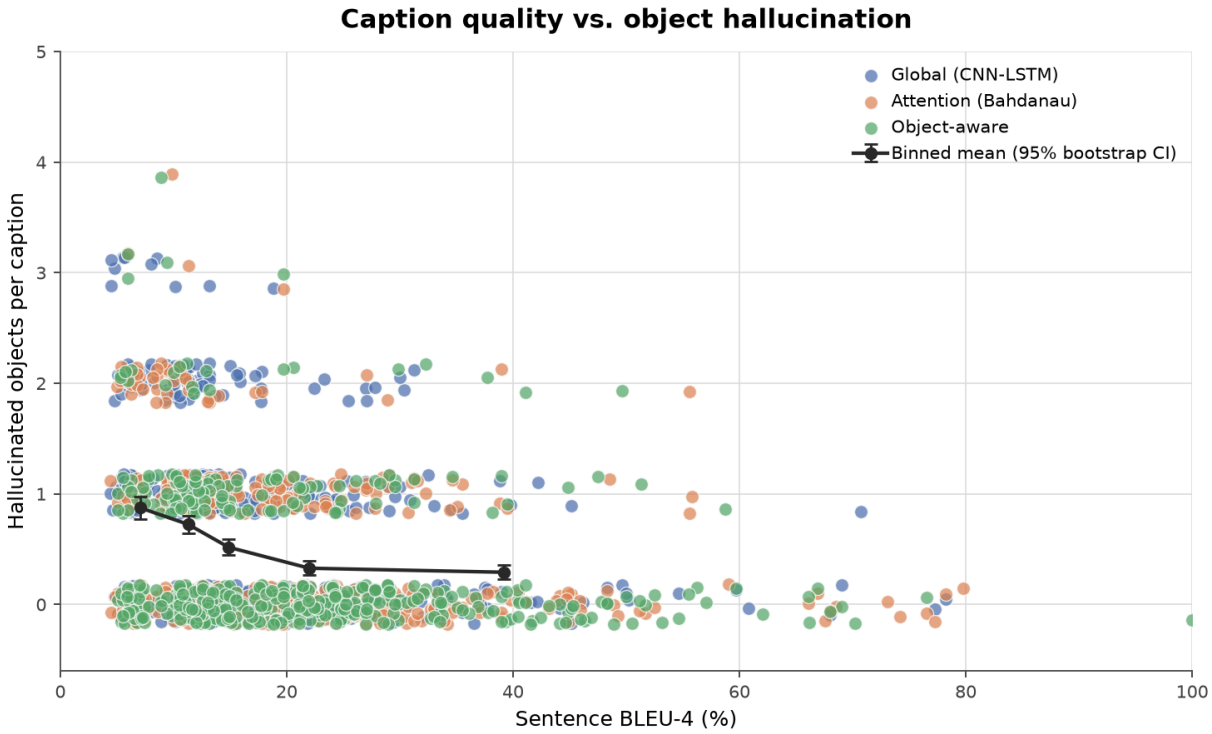
Comparison (first − second)	Δ objects [95% CI]	p	Δ rate (pp) [95% CI]	Δ BLEU-4 [95% CI]	McNemar p
Global vs Attention	+0.30 [+0.23, +0.37]	< 0.001	+21.8 [+17.0, +26.8]	-3.27 [-4.18, -2.41]	< 0.001
Global vs Object-aware	+0.46 [+0.39, +0.53]	< 0.001	+34.4 [+29.6, +39.0]	-4.80 [-5.71, -3.85]	< 0.001
Attention vs Object-aware	+0.16 [+0.11, +0.22]	< 0.001	+12.6 [+8.2, +17.0]	-1.53 [-2.47, -0.60]	< 0.001

Relative to the global baseline, spatial attention reduced hallucinated objects per image by 0.30 (95% CI 0.23 to 0.37; $p < 0.001$) and improved sentence BLEU-4 by 3.27 points (95% CI 2.41 to 4.18; $p < 0.001$). Adding detected object labels to the attention model reduced hallucinated objects per image by 0.16 (95% CI 0.11 to 0.22; $p < 0.001$) and improved sentence BLEU-4 by 1.53 points (95% CI 0.60 to 2.47; $p < 0.001$). Negative differences mean the first model hallucinates fewer objects. In this run the BLEU differences also favour the better-grounded model, but the effect sizes are not interchangeable, and the matched-BLEU analysis below shows that equal BLEU does not imply equal hallucination.

6.2 What the object labels change

The object-aware model is not simply “the attention model plus a better feature”. It receives detector outputs, which include false positives and missed objects. If the labels are helpful, hallucinated *object mentions* should fall; if the detector is noisy, the model may copy detector mistakes. The measured effect is reported above, and the qualitative grid shows both successes and failures. This is a deliberately realistic test of “detected object labels as input” rather than an oracle-label experiment.

7. Does better BLEU mean fewer hallucinations?



Sentence BLEU-4 versus number of hallucinated objects for all test captions, with a binned trend and bootstrap intervals.

Set	N	ρ (BLEU-4 vs #halluc.)	p	ρ (BLEU-2 vs any halluc.)	p	Halluc. rate (%)
Global	500	-0.33	< 0.001	-0.31	< 0.001	62.00
Attention	500	-0.28	< 0.001	-0.25	< 0.001	40.20
Object-aware	500	-0.25	< 0.001	-0.26	< 0.001	27.60
Pooled	1500	-0.33	< 0.001	-0.31	< 0.001	43.27

The pooled caption-level association is a moderate **negative** correlation (Spearman ρ = -0.327; image-cluster p < 0.001): captions with higher sentence BLEU-4 tend to contain slightly fewer hallucinated objects. Grouping all 1,500 pooled captions into BLEU-4 quartiles, the lowest-BLEU quartile has a hallucination rate of 62.67%, versus 24.00% in the highest-BLEU quartile. Because the pooled rows repeat each image once per model, their naive p-values are descriptive. A cluster bootstrap that resamples the 500 images (1,000 draws) gives ρ = -0.327 (95% CI -0.386 to -0.268; p < 0.001) for BLEU-4 versus hallucinated objects, so the association survives the clustering correction.

BLEU-4 quartile	N	BLEU-4 range	Mean halluc. objects	Halluc. rate (%)	95% CI
Q1	375	4.41–10.55	0.87	62.67	57.9–66.9
Q2	375	10.55–14.58	0.63	51.73	46.7–56.8
Q3	375	14.58–24.09	0.40	34.67	30.1–39.7
Q4	375	24.09–100.00	0.28	24.00	19.5–28.3

The model-level rank correlation is -1.00 (higher BLEU goes with lower CHAIRs); with only three operating points it is descriptive, not a significance test. The more useful signal is the caption-level analysis, where 1,500 captions

(500 images × 3 models) provide within-model variation. A negative correlation means that, on average, captions that overlap the references more also mention fewer absent objects. It does not mean that BLEU is sufficient: at matched BLEU levels the models still differ substantially in hallucination (Section 7.1), and the caption-level correlation is only moderate.

7.1 Is BLEU sufficient at matched quality?

A single global correlation can hide a simple confound: models with better language modelling produce both higher BLEU and fewer hallucinations because they produce better captions overall. To separate the two, we compare models inside 10-point sentence-BLEU bins. If BLEU fully explained hallucination, the three models would have similar rates within each bin.

BLEU-4 bin	Model	N	Mean BLEU-4	Halluc. rate (%)	95% CI
[0, 10)	Global	127	7.0	77.2	69.3–84.3
[0, 10)	Attention	108	7.4	57.4	47.2–66.7
[0, 10)	Object-aware	80	7.2	43.8	32.5–53.8
[10, 20)	Global	251	13.4	66.1	60.6–72.1
[10, 20)	Attention	209	14.2	43.1	36.4–50.3
[10, 20)	Object-aware	206	14.2	32.5	26.2–38.8
[20, 30)	Global	76	24.4	44.7	34.2–56.6
[20, 30)	Attention	105	24.5	30.5	21.9–39.0
[20, 30)	Object-aware	127	24.7	17.3	11.0–23.6
[30, 40)	Global	26	34.7	34.6	15.4–53.8
[30, 40)	Attention	45	34.1	28.9	17.8–42.2
[30, 40)	Object-aware	38	35.0	21.1	10.5–34.2
[40, 101)	Global	20	54.4	15.0	0.0–30.0
[40, 101)	Attention	33	56.3	12.1	3.0–24.2
[40, 101)	Object-aware	49	51.9	12.2	4.1–22.4

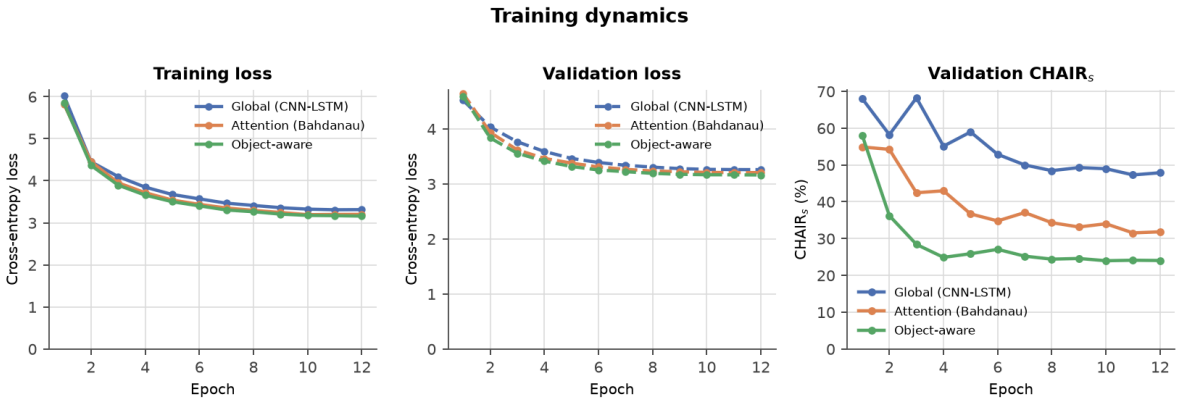
This is the strongest evidence against using BLEU as a grounding metric. In the 10–20 BLEU bin, for example, hallucination rates differ substantially across models even though their mean BLEU is nearly identical. The object-aware model is consistently better in the low- and mid-BLEU bins; the differences shrink at the very highest BLEU, where all models produce conservative, frequently reference-like captions. BLEU therefore captures part of caption quality, but it does not identify which model is grounded.

7.2 Learning dynamics

Training curves are shown below. CHAIRs is computed on the 500 validation images after every epoch; BLEU is reported at final test time (the early checkpoints are not retained). The validation-loss/CHAIR association is a sanity check that better-fitting models tend to hallucinate less within a run.

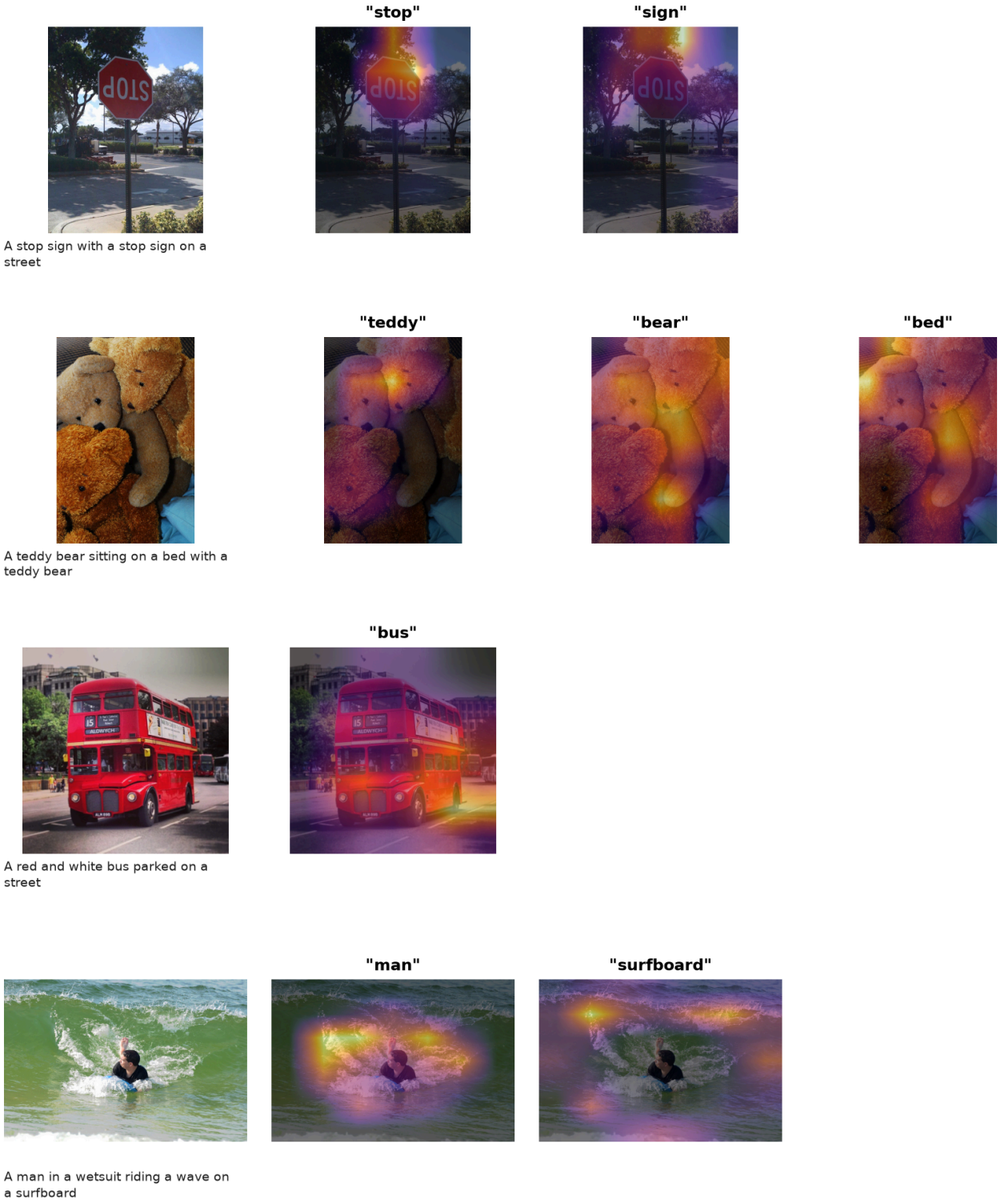
Model	Epochs	Train loss	Val loss	Val CHAIRs (%)	Train time (min)
Global	12	3.314	3.259	47.87	6.9
Attention	12	3.202	3.206	31.84	16.3
Object-aware	12	3.160	3.162	24.03	10.9

Model	Epochs	ρ (val loss vs CHAIRs)	p
Global	12	0.92	< 0.001
Attention	12	0.96	< 0.001
Object-aware	12	0.92	< 0.001



Training and validation curves. The hallucination rate is tracked on the validation split each epoch.

8. Attention and error analysis



Word-by-word attention maps for four held-out images. Each column after the image shows the attention distribution when the highlighted word was generated.

The attention maps show whether the decoder looks at plausible regions when it names an object. Strong, localised maps for nouns are a necessary but not sufficient condition for grounding: a model can attend to the right region and still choose the wrong object word.



GT objects: microwave, teddy bear

Global (CNN-LSTM)
A **cat** sitting on a **bed** with a **laptop** and a **laptop**
Hallucinated: bed, cat, laptop

Attention (Bahdanau)
A **cat** sitting on a desk with a **laptop**
Hallucinated: cat, laptop

Object-aware
A **cat** is sitting on a **table** next to a **laptop**
Hallucinated: cat, dining table, laptop

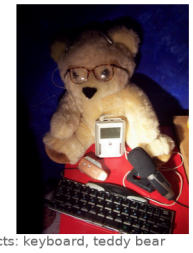


GT objects: bed, book

Global (CNN-LSTM)
A man and a woman sitting on a **table** with a **cake**
Hallucinated: cake, dining table, person

Attention (Bahdanau)
A **person** holding a video game of **cake**
Hallucinated: cake, person

Object-aware
A **person** is sitting on a bed with a **laptop**
Hallucinated: laptop, person



GT objects: keyboard, teddy bear

Global (CNN-LSTM)
A **cat** sitting on a **bed** with a **laptop** computer
Hallucinated: bed, cat, laptop

Attention (Bahdanau)
A **cat** is sitting on a desk with a **laptop**
Hallucinated: cat, laptop

Object-aware
A man is sitting on a desk with a **laptop**
Hallucinated: laptop, person



GT objects: vase

Global (CNN-LSTM)
A **bowl** of food with a bunch of bananas
Hallucinated: banana, bowl

Attention (Bahdanau)
A **bowl** of food with a red and white **cake**
Hallucinated: bowl, cake

Object-aware
A **bowl** of food with a **glass** of flowers
Hallucinated: bowl, wine glass

Qualitative comparison. Hallucinated object words are highlighted in red.

The qualitative grid is intentionally selected to show disagreement between models, so it is not a random sample. It illustrates the two failure modes that matter for this assignment: (i) a fluent caption that mentions an absent object, and (ii) a conservative caption that avoids hallucination at the cost of detail or BLEU.

9. Discussion

The experiment separates three effects that are often conflated:

1. **Attention is not automatically grounding.** Attention gives the decoder a spatial interface, but the training signal only rewards word likelihood. A model can learn useful attention maps and still over-generate familiar objects.

2. **Object labels are a strong but noisy prior.** Detector labels make object vocabulary available at every decoding step. This can reduce hallucination, but false positives and missed detections propagate into the caption.
3. **BLEU is a partial proxy.** Better captions usually mention more of the right content, so BLEU and hallucination are negatively correlated at the caption level. The relationship is only moderate, and at matched BLEU the models still differ substantially in hallucination. BLEU therefore captures part of caption quality but does not identify which model is grounded.

10. Limitations

- **Small training set.** The models train on 4,000 images (20k captions), far below standard COCO training sets. Absolute BLEU/CIDEr values are therefore low; the comparison is controlled but not state of the art.
- **COCO val re-split.** The official validation images are re-partitioned. This is a laptop-scale compromise; results are not directly comparable to papers that train on train2017.
- **CHAIR annotation incompleteness.** An object that is visible but missing from COCO annotations is counted as a hallucination. This inflates CHAIR for all models, especially for small or uncommon objects. Conversely, ambiguous synonyms (for example “glass”, “bat” or “mouse”) can be mapped to a COCO class even when the caption means something else.
- **Detector noise.** The object-aware model receives detector outputs, not ground-truth labels. That is realistic but makes the treatment noisy.
- **Duplicate caption strings.** COCO contains repeated caption strings across different images; 27 test strings also occur in training on different images. There is no image-ID overlap, and excluding them changes BLEU-4 by less than 0.05 in this run.
- **Greedy decoding and one run.** Results use greedy decoding and a single seed; beam search and multiple seeds would reduce variance.
- **No human evaluation.** Fluency and correctness are assessed automatically.

11. Conclusion

We built a reproducible COCO captioning benchmark that measures object hallucination alongside BLEU and CIDEr-D. The three controlled models show that attention and detected object labels improve both caption quality and grounding. The BLEU leader (Object-aware) was also the model with the lowest object-hallucination rate (23.64%). The pooled caption-level association is a moderate **negative** correlation (Spearman $\rho = -0.327$; image-cluster $p < 0.001$): captions with higher sentence BLEU-4 tend to contain slightly fewer hallucinated objects. The caption-level association is only moderate, and at matched BLEU the models still differ substantially in hallucination. The practical recommendation is simple: report a grounding metric such as CHAIR next to BLEU, and do not use BLEU alone to choose a captioning model.

References

1. Vinyals, O., Toshev, A., Bengio, S., & Erhan, D. (2015). *Show and Tell: A Neural Image Caption Generator.* CVPR. arXiv:1411.4555.

2. Xu, K., Ba, J., Kiros, R., Cho, K., Courville, A., Salakhutdinov, R., Zemel, R., & Bengio, Y. (2015). *Show, Attend and Tell: Neural Image Caption Generation with Visual Attention.* ICML. arXiv:1502.03044.
3. Rohrbach, A., Hendricks, L. A., Burns, K., Darrell, T., & Saenko, K. (2018). *Object Hallucination in Image Captioning.* EMNLP. arXiv:1809.02156.
4. Anderson, P., He, X., Buehler, C., Teney, D., Johnson, M., Gould, S., & Zhang, L. (2018). *Bottom-Up and Top-Down Attention for Image Captioning and Visual Question Answering.* CVPR. arXiv:1707.07998.
5. Vedantam, R., Zitnick, C. L., & Parikh, D. (2015). *CIDEr: Consensus-based Image Description Evaluation.* CVPR. arXiv:1411.5726.
6. Papineni, K., Roukos, S., Ward, T., & Zhu, W.-J. (2002). *BLEU: a Method for Automatic Evaluation of Machine Translation.* ACL.
7. Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). *Microsoft COCO: Common Objects in Context.* ECCV. arXiv:1405.0312.
8. Ren, S., He, K., Girshick, R., & Sun, J. (2015). *Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks.* NeurIPS. arXiv:1506.01497.
9. He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep Residual Learning for Image Recognition.* CVPR. arXiv:1512.03385.

Appendix: Reproducibility

Item	Value
Backbone	ResNet-18, ImageNet weights, `layer4` 7×7×512
Feature extraction	60.2 s for 5,000 images
Detector	fasterrcnn_mobilenet_v3_large_320_fpn
Detection time	13.0 min
Object threshold	0.3
Max detections	12
Optimiser	AdamW, OneCycle, lr=0.001
Epochs	12
Sampled pairs / epoch	8,000
Hardware	Apple M4, 16 GB unified memory, macOS
Device	CPU for LSTM training; MPS for feature/detector extraction
MPS memory cap	55%

All metrics in this report are generated by `scripts/05_evaluate.py`; the figures are generated by `scripts/06_visualize.py`. Run `scripts/07_make_report.py` to rebuild this document.